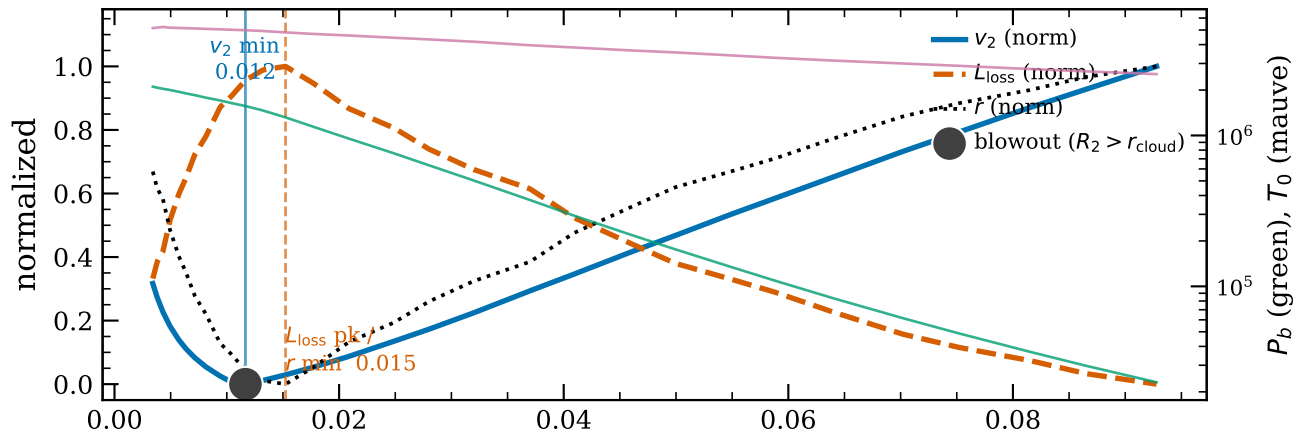
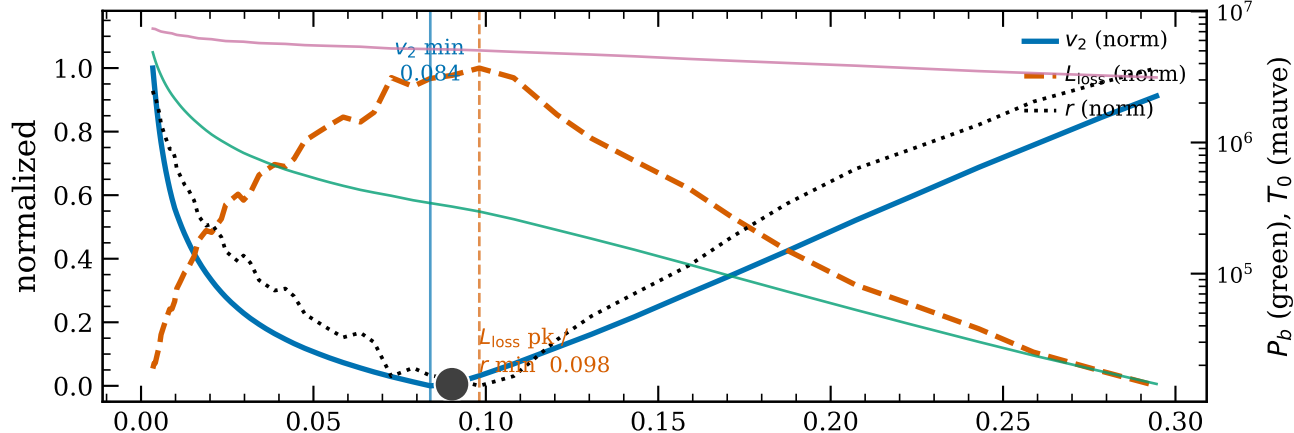


Causal order through the early cooling-ratio dip: v_2 decelerates first, L_{loss} peaks at/after, dE_b/dt stays positive (no sign change) while P_b, T_0 fall monotonically

small dense, high SFE: $v_2 \text{min}=0.012$, $L_{\text{loss}} \text{pk}=0.015$ (v_2 LEADS); dE_b/dt no sign change



simple cluster (baseline): $v_2 \text{min}=0.084$, $L_{\text{loss}} \text{pk}=0.098$ (v_2 LEADS); dE_b/dt no sign change



midrange, flat profile: $v_2 \text{min}=0.392$, $L_{\text{loss}} \text{pk}=0.482$ (v_2 LEADS); dE_b/dt no sign change

